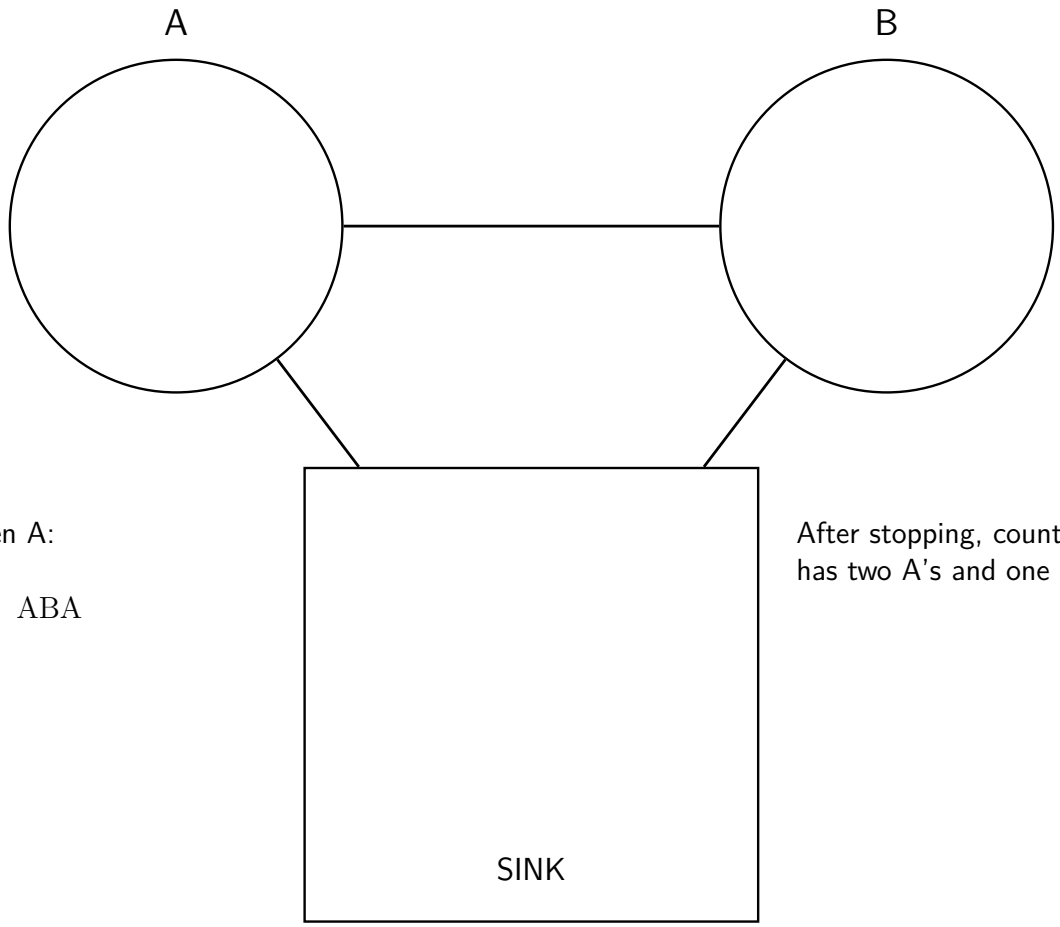
A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start. Write each sharing letter in order, then the final piles.

Problem 1: Finish each start in two different orders, if possible. After stopping, count each letter in the sharing word. For the same start, can the final piles agree while the sharing counts differ?



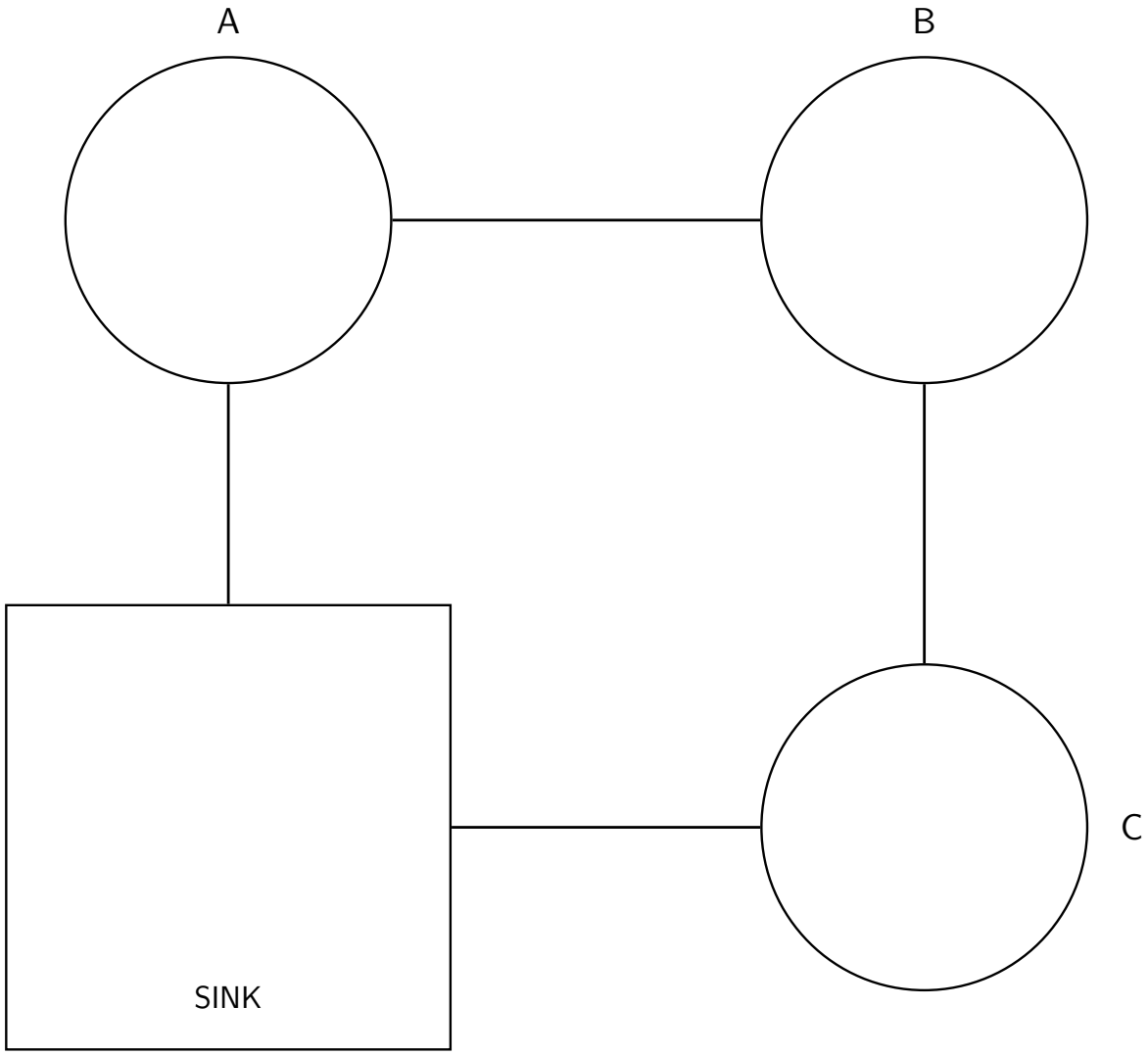
A, then B, then A:

$\mathrm{A} \rightarrow \mathrm{AB} \rightarrow \mathrm{ABA}$

After stopping, count letters: ABA has two A's and one B.

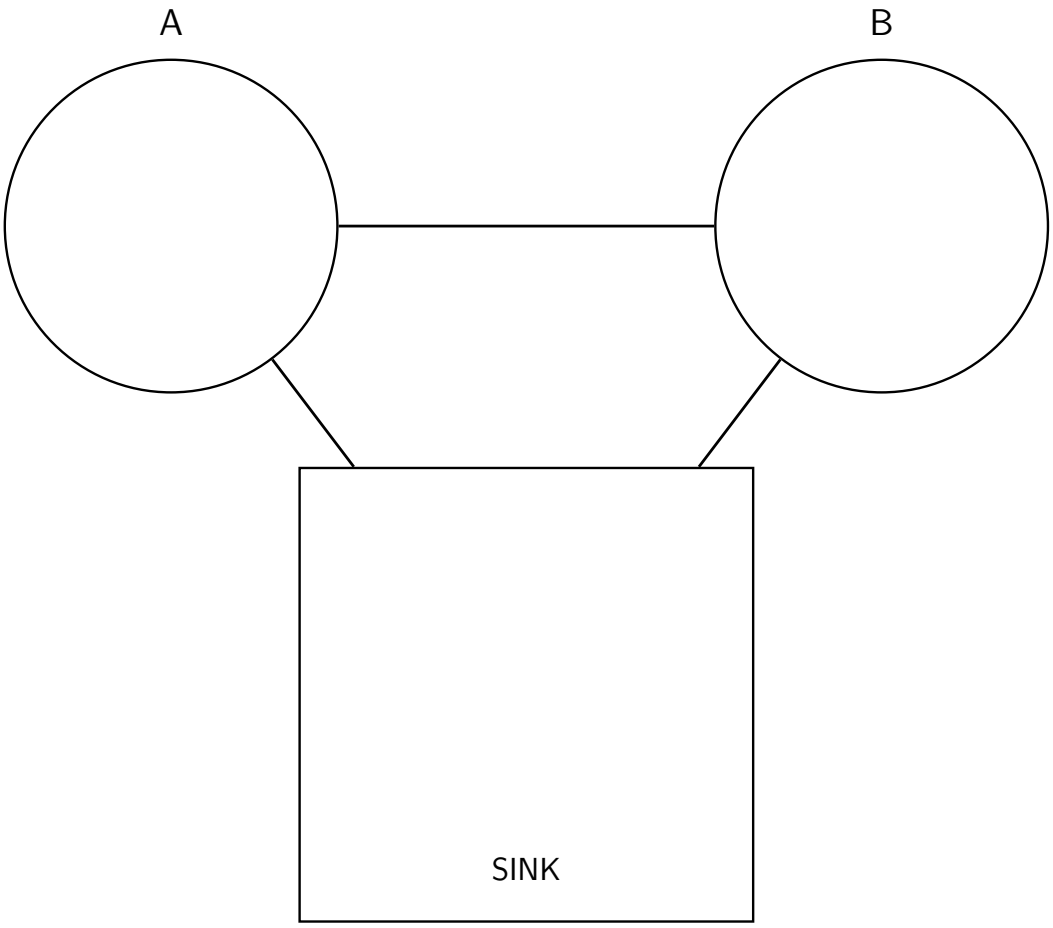
Start A	Start B	Sharing word	Finish A	Finish B
2	2			
2	2			
4	0			
4	0			
3	3			
3	3			
5	4			
5	4			

Problem 2: Start empty. For each pair of batches, compare adding them together with finishing one before adding the other. Try both batch orders. Does the timing of additions change the finish?



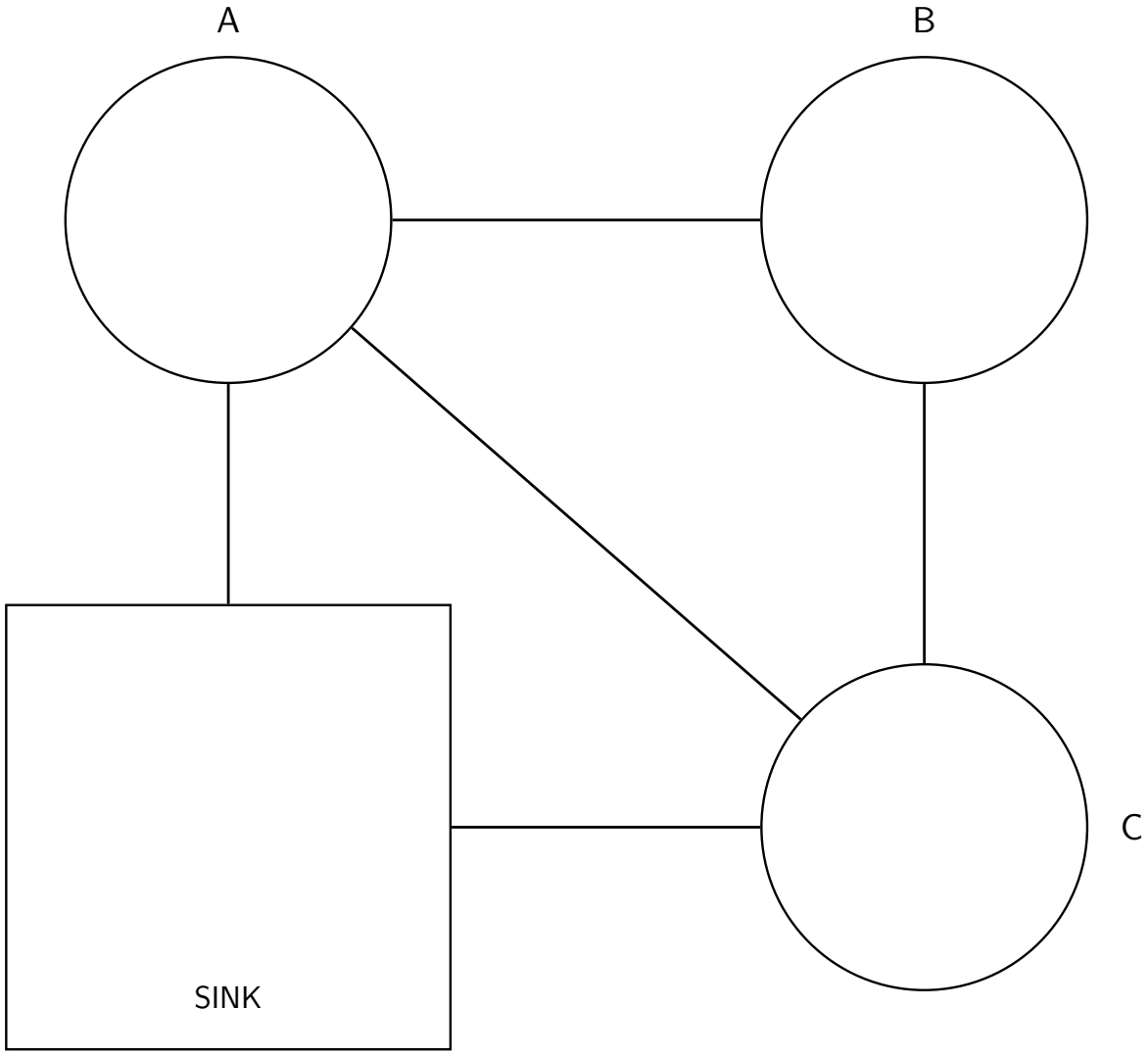
First batch	Second batch	Timing	Finish A	Finish B	Finish C
4 at B	2 at A	together			
4 at B	2 at A	first, then second			
4 at B	2 at A	second, then first			
3 at A	3 at C	together			
3 at A	3 at C	first, then second			
3 at A	3 at C	second, then first			

Problem 3: From each start, repeatedly add one chip to A and finish after each addition. Record the pairs you get. Which starts return to themselves? Can two different starts give the same pair after one addition?



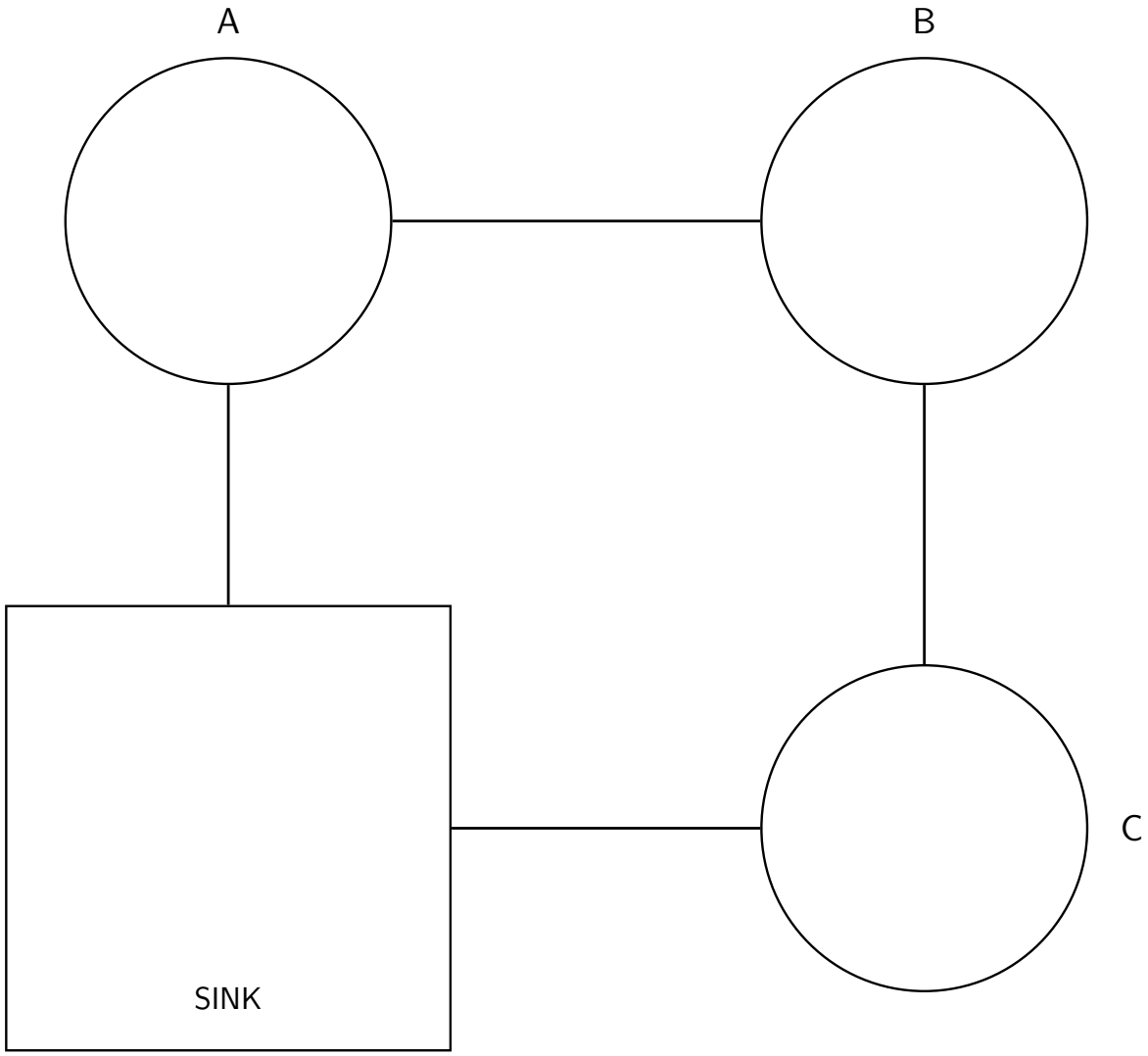
Start A	Start B	After 1	After 2	After 3	After 4
0	0				
1	0				
0	1				
1	1				

Problem 4: Finish each start in two different orders, if possible. After stopping, compare the final piles and count each letter in the sharing words. Does the extra line change your conclusion about order?



Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
3	0	3				
3	0	3				
0	6	0				
0	6	0				
4	1	2				
4	1	2				

Problem 5: Try starts with all 12 chips at B, with 6 at A and 6 at C, and with 4 at each circle. Could any order of sharing continue forever? Explain why sharing must stop for any finite number of chips on this board.



Problem 6: Start with 8 chips at B and none at A or C. Find two different orders that finish. Explain why every order must give the same final piles and sharing counts, for this start and for any other start on this board.

